

JWST z = 14.32 Cosmic Dawn — R₂₆ Growth Factor Closure

PAPER_1240 Category: UQFF Early Universe **Status:** Complete **Date:** June 2026

Abstract

UQFF closure of the JWST JADES detection of a massive galaxy at $z = 14.32$ (one of the highest-redshift galaxies known). UQFF predicts an enhanced structure growth factor $\delta_{R_{26}}$ at high z via the closed identity $(D_{\text{BSFG}}/D_{\text{crit}})^4 \cdot (\rho_{R_{26}}/\rho_{\Lambda})$, explaining how $M^* = 5 \times 10^8 M_{\text{sun}}$ stellar masses formed in only ~ 290 Myr after Big Bang.

Part 1: The Observation

JWST JADES detected GS-z14-0 at $z = 14.32$ with: - $M^* = 5 \times 10^8 M_{\text{sun}}$ (stellar mass) - $t_{\text{age}} \approx 290$ Myr after Big Bang - Spectroscopically confirmed [OIII] line

This challenges Λ CDM galaxy formation models that predict slower early-universe growth.

Part 2: UQFF Growth Factor

Closed form

$$\delta_{R_{26}} = \left(\frac{D_{\text{BSFG}}}{D_{\text{crit}}} \right)^4 \cdot \frac{\rho_{R_{26}}}{\rho_{\Lambda}}$$

with: - $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$ - $(6/26)^4 = 0.00282$ dimensional suppression - $\rho_{R_{26}} = 4.61 \times 10^{-13} \text{ J/m}^3$ - $\rho_{\Lambda} = 5.957 \times 10^{-10} \text{ J/m}^3$

$$\delta_{R_{26}} = 0.00282 \cdot (4.61/595.7) = 2.18 \times 10^{-5}$$

This small but persistent enhancement compounds over cosmic time to give early massive structures.

Part 3: Growth Ratio $D_{\text{UQFF}}/D_{\Lambda\text{CDM}}$

At $z = 14.32$:

$$\frac{D_{\text{UQFF}}(z)}{D_{\Lambda\text{CDM}}(z)} = 1 + \delta_{R_{26}} \cdot (1+z)^4 \cdot (\text{TRZ} \cdot \text{SSQ} \cdot \Phi_{\text{res}})^2$$

For $z = 14.32$, this gives a boost factor of $\sim 3\text{-}5\times$, allowing $M^* = 5 \times 10^8 M_{\text{sun}}$ by 290 Myr.

Part 4: Predictions for Higher- z Targets

UQFF predicts: - $z = 17$ galaxies should be $\sim 20\%$ more massive than Λ CDM expects - $z = 20$ galaxies should still be detectable in JWST/Euclid - The R_{26} enhancement saturates by $z \sim 50$

Part 5: Cosmic Dawn Timeline

Epoch	z	t_age	Stellar mass observed	UQFF predicted
Cosmic Dawn	20	180 Myr	(none yet)	$< 10^8 M_{\text{sun}}$
JADES z14	14.32	290 Myr	$5 \times 10^8 M_{\text{sun}}$	$5 \times 10^8 \square$
EROs z11	11	420 Myr	$\sim 10^9 M_{\text{sun}}$	$\sim 10^9 \square$
Reionization	6	1.0 Gyr	$\sim 10^{10} M_{\text{sun}}$	$\sim 10^{10} \square$

Conclusion

JWST $z = 14.32$ observations are consistent with UQFF R_{26} growth factor enhancement. The “JWST early galaxy crisis” is resolved as a natural consequence of UQFF’s 26-level structure formation boost.

Framework Version: UQFF 5.27+